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“Sorry, it was my fault”: Repairing trust in human-robot interactions

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ABSTRACT

The current study develops a three-fold categorization (i.e., logic, semantic, and syntax failures) for technical failures that are commonly observed in human-robot interactions (HRI), and investigates it along with four trust repair strategies: internal-attribution apology, external-attribution apology, denial, and no repair. The 743 observations conducted through an online experiment reveal some nuances in participants' perceprions of competence- and integrity-based trust violations, which may reflect ontological differences between humans and machines. The analysis also shows significant main effects of failure types and repair methods on HRI-based trust.

1. Introduction

As uses of technology become more deeply interwoven with human life, the relationships between humans and technology have grown more interdependent (Guzman and Lewis, 2020). Consequently, trust is no longer a socio-psychological concept only applicable to interpersonal dynamics. Akin to trust developed through human-to-human communication, trust toward technology also reflects trustors' evaluations of trustees' (i.e., machines) abilities and helpfulness in achieving expected goals. Since trust is closely associated with interaction outcomes and usage decisions (Sanders et al., 2019), trust in human-machine communication (HMC) has attracted significant scholarly interest. In this paper, HMC is an umbrella term indicating all types of communication between humans and machines including computers, automations, artificial intelligence, and robots.

The present study specifically focuses on trust in robotics. Following Lee and See (2004), the current study defines trust as “the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability” (p. 54). As illustrated by Hoff and Bashir (2015), this kind of trust has three fundamental layers: dispositional trust (i.e., trust based on individuals' enduring trust tendency), situational trust (i.e., trust based on a specific context of interaction), and learned trust (i.e., trust based on past experience with a specific system). Distinct from interpersonal trust, trust in robots is characterized by a heavier emphasis on system performance (Baker et al., 2018; Hancock et al., 2011), which directly impacts learned trust. Since numerous errors can occur in human-robot interactions, leading to declines of trust (Desai et al., 2012; 2013; Salem et al., 2015), it becomes particularly important for roboticists to enhance robots' capabilities for detecting performance failures and repairing users' trust (Brooks, 2017; Sebo et al., 2019).

Humans employ various strategies to rehabilitate interpersonal trust, and these strategies could potentially be transplanted to the context of human-robot interaction (HRI) (de Visser et al., 2018). The current study first identified three types of technical failures resulting from basic system errors (i.e., logic, semantic, and syntax errors; McCall and Kölling, 2014) and revised the human-automation trust repair framework proposed by Marinaccio et al. (2015). Next, the present study explored the effectiveness of three trust recovery tactics (i.e., apology with internal or external attributions and denial), considering no repair as a reference point, and investigated the potential interaction between failure types and repair methods.

2. Trust violations and repairs

As a critical factor for development and maintenance of social and professional relationships (Haesevoets et al., 2015), trust has long been a topic of intense research across various disciplines (Lewicki and Brinsfield, 2017; Schaefer, 2013). Development and maintenance of trust is inevitably accompanied by risks of trust violations, defined as “unmet expectations concerning another's behavior, or when the person does not act consistent with one's values” (Bies and Tripp, 1996, p.248)

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resulting in social and economic loss (Rao and Lee, 2007). In response to perceived transgressions, trustees can attempt to obtain forgiveness (Tomlinson et al., 2004) and restore positive expectations while minimizing negative ones (Kramer and Lewicki, 2010). In contrast to the abundance of trust literature, heretofore research on repair strategies has been a late bloomer (Dirks et al., 2009). In the field of HMC, trust repair is still an understudied subject that seeks theoretical support from human-to-human trust literature, although the dynamics between the two may be substantially different. Thus, we choose to review relevant literature in both areas.

2.1. Human-to-human trust repair

The current trust repair studies in HMC are mainly inspired by a burgeoning line of trust repair research in the field of organizational communication. A series of studies (Ferrin et al., 2007; Kim et al., 2004, 2006, 2013) primarily investigated two repair strategies: (a) apology (i.e., a response in which trustee accepts responsibility, expresses repentance, and stresses the intent to avoid similar violations in the future) and (b) denial (i.e., a response in which the trustee rejects responsibility and expresses no repentance); under two types of trust violations: (a) competence-based violations and (b) integrity-based violations. Competence/Ability (i.e., "group of skills, competencies, and characteristics that enable the party to have influences within some specific domain", p.717) and integrity ("the trustor's perception that the trustee adheres to a set of principles that the trustor finds acceptable", p.719) are critical determinants of trust (Mayer et al., 1995), and the two are fundamentally different because they invoke different attribution patterns—competence and performance can be tested, but integrity and morality cannot be easily quantified (Reeder and Brewer, 1979).

Consequently, Kim et al. (2004) assumed that apology is more effective with competence-based violations than denial, since trustors pay more attention to positivity (i.e., expressing repentance and making promises) brought by acknowledgement of such violations rather than negativity, By contrast, denial is a more potent succor for integrity-based violations, because avoiding negativity (i.e., admitting lack of integrity) regarding such violations should be more effective than generating positivity. Initially, Kim et al. (2004) observed significant interactions between the two violation types and two repair methods on trusting beliefs and intentions. Subsequent studies mostly substantiated their preliminary findings (Ferrin et al., 2007; Kim et al., 2006; 2013). In the recent decade, trust repair research has proliferated in the field of organizational communication (Bachmann et al., 2015; Eberl et al., 2015; Fuoli et al., 2017; Gillespie and Dietz, 2009; Janowicz-Panjaitan and Krishnan, 2009; Poppo and Schepker, 2010), exploring this subject on multiple organizational levels.

Notwithstanding the notable support Kim et al.'s (2004) model has received, a few studies have indicated otherwise. The study by Utz et al. (2009) showed that a plain apology was considered more believable than a denial for both competence- and integrity-based violations by eBay buyers. Bansal and Zahedi (2015) argued that apology was superior to denial for every violation type in a scenario of privacy breach, and denial performed even worse than no response under some conditions. Such counterevidence denotes that the model established by Kim et al. (2004) may not be equally applicable under certain contexts, since the mechanisms underlying apology and denial are highly complex.

Apart from the interactions depicted by Kim et al. (2004), scholars have also explored other key elements affecting reconciliation between trustors and trustees. Prior studies have highlighted timing, severity, and frequency of violations (Lewicki and Brinsfield, 2017), dispositional trust (Colquitt et al., 2007; Kramer, 1999), relationship characteristics (e.g., relative status; Aquino et al., 2001; past relationships and probability of future violations; Tomlinson et al., 2004) to be influential factors in trust repair. Repair intensity, perceived sincerity, and multi-dimensionality (i.e., display of regret, explanations, acknowledgement of accountability, offer of future repair, and entreaty for forgiveness) of apology (Lewicki and Brinsfield, 2017; Lewicki and Tomlinson 2003; Tomlinson et al., 2004), as well as timeliness of act, are some additional variables associated with repair effectiveness. Trustees' characteristics also matter, including personal traits (e.g., likeability; Bradfield and Aquino, 1999; gender; Walfisch et al., 2013), and organizational features (e.g., pre-crisis reputations; Beldad et al., 2018; Lewicki and Tomlinson, 2003). To summarize, trust repair is a highly complex practice because of the multifaceted nature of trust and variance of repair context, which invites further theoretical and empirical exploration.

2.2. Robot-to-human trust repair

Research focusing on trust promotion in HMC is copious, but research into trust in robotic systems is a relatively novel emphasis (Baker et al., 2018). Particularly, existing trust literature in HMC sheds light on technical designs (e.g., visual anthropomorphism, machine politeness) that increase baseline trust levels and hence benefit trust resilience (de Visser et al., 2016; Quinn, 2018). In the context of HRI, factors affecting trust development can be roughly classified into human-related (ability-based factors and characteristics), robot-related (performance- and attribute-based factors), and environment/context-related factors (task and team collaboration), according to Hancock and his colleagues (2011), out of which performance-based factors turn out to be the central determinant in human evaluations of robotics.

However, performance failures are ubiquitous in HRI, after which human-to-machine trust naturally declines (Corritore et al., 2003; Desai et al., 2012, 2013; Salem et al., 2015; Sanchez et al., 2014; de Vries et al., 2003). In response, prior research suggests that robots can initiate trust repair just as humans can, bolstering perceived sociability and humanness while further promoting trust resilience (de Visser et al., 2012, 2016, 2018), and such attempts at trust repair may remain effective even when machines' reliability does not really improve (de Visser, 2012). For trust repair in HMC, previous findings encompassed various repair strategies, such as ignoring (Correia et al., 2018), blaming (Groom et al., 2010; Kaniarasu and Steinfeld, 2014), apologizing (de Visser et al., 2016; Lee et al., 2010; Quinn, 2018; Robinette et al., 2015; Sebo et al., 2019; Tzeng, 2004; Wagner, 2016), denying (Quinn, 2018; Sebo et al., 2019), promising (Robinette et al., 2015; Wagner, 2016), justifying (Correia et al., 2018), engaging in social dialogues (Lucas et al., 2018), giving palpable compensations (Lee et al., 2010), offering options (Lee et al., 2010), and providing additional information (Robinette et al., 2015) or support (Brooks, 2017). In general, these studies espouse the perspective that it is better to take repair actions than not, but the investigation has been fragmented, lacking systematic approaches (de Visser et al., 2020).

Most importantly, there have been two studies examining Kim et al.'s (2004) framework in the context of HMC. Quinn (2018) found that although apology was more effective as a repair method for competence-based violations than for integrity-based violations, repair outcomes of denial did not significantly differ for the two types of violations. Sebo et al. (2019) substantiated the interactions identified by Kim et al. (2004) with a competitive shooting game in which a robot broke its initial promises and framed the reasons to be either competence- or integrity-based. The researchers noted that human players were more inclined to retaliate under the condition of integrity-based violations plus denial. These findings verify that the conclusions about trust repair in human-to-human communication remain instructive in HMC, although the fundamental differences between humans and robots have not been subjected to rigorous consideration.

3. Uniqueness of HRI-based trust

3.1. Perceptual differences

Existing trust research of HMC has exhibited some commonality bridging human-to-human and human-to-machine communication. Nevertheless, a few studies have unveiled some key differences between interpersonal trust and HMC-based trust. Humans seem to possess different levels of dispositional trust toward humans and machines, allocating more initial trust to the latter owing to a positive bias toward automation (de Visser et al., 2016; Dzindolet et al., 2003; Hoff and Bashir, 2015; Madhavan and Wiegmann, 2005; Parasuraman and Manzey, 2010). As a corollary, people may overreact to system failures due to interruptions of perfect automation schema (Dzindolet et al., 2002; Madhavan and Wiegmann, 2005). Therefore, trust in machines can be harder to reestablish once violated than interpersonal trust (Hoffman et al., 2013), as machine failures can potentially lead to greater negative expectation violations.

Conceptual differences between humans and technologies also add to the perceptual inequivalence toward trust violations and their repairs. First and foremost, the belief that machines are more fixed than humans can potentially impair the effects of trust repair, because users may hold the opinion that an oral repair is not likely to be followed by an actual improvement in machines' performance (de Visser et al., 2018). They may perceive trust repair efforts from machines to be less sincere because repair is predefined by algorithms (de Visser et al., 2018), especially when repair attempts appear uniform across different kinds of situations. Second, humans also perceive morality in machines differently because machines do not have human minds, which means they cannot accumulate sensational experiences as humans do (Banks, 2019; Gray et al., 2012). Thus, human judgments of machines related to moral principles (i.e., integrity and benevolence) may differ from their judgments of other humans.

Additionally, machines are also perceived to possess less agency and are viewed as less legitimate in making moral decisions (Gray et al., 2007, 2012; Malle et al., 2016). As posited by Parasuraman and Riley (1997), users may feel they are building trust with designers instead of machines during interactions, so it is also probable that humans perceive morality-related violations differently and make different violation attributions during HRI. Sebo et al. (2019) manipulated their robot to explicitly articulate the reasons (e.g., "Oh no! I hit the wrong button" and "Yes! You're immobilized") of trust violations to ensure that interactants perceived them as certain kinds of violations. Since, however, machines do not frame the intentions behind violations so clearly in real life, it remains uncertain whether interactants in HMC perceive competence- or integrity-based violations in the same manner as they do in interpersonal communication.

3.2. Robotic failures

"Failure" and "error" are often used interchangeably in HMC research together with "fault". In the present study, they are approached as overlapping but distinctive terms. First, the term "failure" refers to "a degraded state of ability which causes the behavior or service being performed by the system to deviate from the ideal, normal, or correct functionality" (Brooks, 2017, p.9), emphasizing violations of interactants' subjective expectations. Second, the word "error" is a more technical term, encompassing "system states (electrical, logical, or mechanical) that can lead to a failure" (Honig and Oron-Gilad, 2018). Third, the term, "fault" is defined as lower-order sources of errors (Honig and Oron-Gilad, 2018). Errors can cause failures, if noticed and perceived by human users as failures, but failures do not necessarily result from errors—misperceptions and incompatible design principles can also engender failures. From the robotic end, failures caused by system errors are competence-based, because they are caused by unintended system inability preventing robots from producing correct output and executing human commands accurately, but they are not necessarily perceived to be competence-based trust violations by human interactants.

3.2.1. Taxonomies of failures

Previously, numerous failure typologies were constructed to explain errors emerging in HMC that are introduced by humans, robots, and the environment. Based on the locus of faults, they were categorized into (1) information acquisition, information analysis, decision/action selection, and action implementation (Parasuraman et al., 2000), (2) interaction, algorithms/methods, software design/implementation, and hardware failures (Carlson and Murphy, 2005), (3) interactions, algorithms, software, and hardware faults (Steinbauer, 2012); and (4) communication failures and processing failures (Brooks, 2017). Based on situations of expectation violations, Giuliani et al. (2015) distinguished failures by technical failures (e.g., a robot dropping something by accident) and social norm violations (e.g., a robot looking away from the interactant during the conversation). Based on mechanisms of errors, Skitka et al. (2000) divided omission and commission errors. Based on combinations of failures, Ferrell (1994) organized robotic failures into individual (i.e., a failure occurs alone), concurrent (i.e., a failure occurs concurrently with other failures), and accumulative (i.e, failures accumulate over time) failures. Based on severity of aftereffect, they were classified into (1) benign failures (e.g., a failure that occurs in a fail-safe system and can be automatically detected, causing no harm) and catastrophic failures (e.g., a failure leading to catastrophic damage) (Laprie, 1995), (2) non-critical, repairable/compensable, and terminal failures (Carlson and Murphy, 2005). Based on recoverability of failures, they could be divided into anticipated (i.e., a failure that can be fixed through alternative options in the original plan), exceptional (i.e., a failure requiring replanning), and unrecoverable errors (i.e., a failure that can be neither fixed through original planning nor replanning) (Ross et al., 2004). Also, based on whether failures are relevant to a specific robotic system or applicable across all systems, they were identified as high, medium, and low relevancy failures (Honig and Oron-Gilad, 2018).

Despite the appreciable amount of effort devoted to taxonomy constructions, these categorizations were rarely integrated into experimental designs, especially in applied research. The error types attracting the greatest amount of scholarly attention so far are commission and omission errors, and this line of research has mainly scrutinized the effects of false-alarms and misses in automation systems. The experiments led to mixed findings: some indicated misses had more negative valence (e.g., Davenport and Bustamante, 2010; Dixon and Wickens, 2003; Sanchez, 2006), but others suggested false alarms were worse (e.g., Johnson et al., 2004), with some viewing both as equally destructive (e.g., Madhavan et al., 2006; Rovira and Parasuraman, 2010). To fill the gap in literature, Hoff and Bashir (2015) commented that the contradictions might have been caused by different consequences of errors across systems. A false alarm of a carbon monoxide detector may simply be an annoyance, yet a miss could lead to casualties. By contrast, false notifications may be considered more negatively for common mobile applications than misses of notifications, which often go unnoticed.

3.2.2. Trust repair based on failure types

Trust repair research related to different violation types is a relatively new topic in HMC, and the framework formulated by Marinaccio et al. (2015) is noteworthy. This framework connects the aforementioned findings from organizational literature (Kim et al., 2004) and human error typology from Reason (1990), surmising that the same interactions between two violation types and two repair methods also manifest themselves in human-automation relationships (see Table 1). Based on the scope of the present study, "violations" and "mistakes" in Reason's classification (1990) were determined to be left out as they are not failures directly resulting from common technical errors.

The same concern pertains to other extant failure taxonomies that assort failures based on systemic locus of faults, as the precise origins of

Table 1
Trust repair framework proposed by Marinaccio et al. (2015).

Error type (Reason, 1990)	Examples	Violation types (Kim et al., 2013)	Effective repair (Kim et al., 2013)
Slips – Errors of commission – when an intended action is wrongly executed	Flipping the wrong switch on an IV pump	Integrity-based	Denial
Lapses – Errors of omission – resulting in failure to carry out the action	Forgetting to administer medication	Competence-based if due to memory failure, integrity-based if due to attention failure	Context-dependent
Mistakes – Errors of planning or judgment	Prescribing an incorrect dosage	Competence-based	Apology
Violations – Intentional commission of an error	Prescribing an inappropriate medication because of sponsor loyalty	Integrity-based	Denial

errors are little known by common users. To resolve such conflicts, basic error types (logic, semantic, and syntax errors; McCall and Kölling, 2014) in computer science can be adopted for developing an execution-centered failure categorization and investigating technical failures in more depth (Table 2).

Take, for instance, a hypothetical task in which a robot is instructed to build a toy tower. Logic errors are termed as errors causing machines to produce relevant but incorrect output, which covers a part of slips, such as retrieving four building blocks when asked to bring three. Semantic errors refer to errors yielding completely irrelevant output inappropriate in the given context, which blankets all non-logic slips, such as singing a song when required to pick up a stick. Syntax errors are essentially errors of omission, in which cases machines fail to run programs and generate no output, such as giving no responses to human commands. The three categories of errors will lead to three types of failures when interactants heed the flaws in robotic output, so they are labeled accordingly as logic, semantic, and syntax failures.

Such failures are common in HRI. For instance, two types of failures naturally occurred in Pino et al.'s (2020) study—their robot sometimes incorrectly evaluated participants' responses and demonstrated logic failures (e.g., judging "10" to be the right sum of "5" and "8") and syntax failures (e.g., not responding to instructions). Command recognition errors also frequently happen to robotics (Iio et al., 2020), which may lead to semantic failures. From automations to virtual systems, this outcome-centered typology transcends the technical divisions of hardware and software and is applicable to most existing systems.

Since humans make moral-related attributions to machines less frequently, we deduct that these failures should all be subjectively

Table 2
Trust repair framework for technical errors only.

Failure types	Examples	Violation types (Kim et al., 2013)	Effective repair (Kim et al., 2013)
Logic failures – resulting in relevant but wrong action	Flipping the wrong switch on an IV pump	Competence-based	Apology
Semantic failures – resulting in irrelevant or meaningless action	Reduce dosage on record when asked to print out a prescription	Competence-based	Apology
Syntax failures – resulting in failures to carry out the action	Forgetting to administer medication	Competence-based	Apology

conceptualized as competence-based violations. This speculation is different from Marinaccio et al.'s (2015) propositions that humans will perceive the robotic motivations behind slips, which include both logic and semantic failures, to be integrity-based. Quinn (2018) and Sebo et al. (2019) have verified in HMC that internal-attribution apology (i.e., admitting its own fault) repairs trust more effectively for competence-based violations than denial, and vice versa for integrity-based violations. If participants perceive all three failure types as competence-based violations, their trust should be better recovered with internal-attribution apology.

H1. After failure occurs, participants' trust in the robot will be repaired more successfully when it repairs trust with internal-attribution apology rather than denial for (a) logic, (b) semantic, and (c) syntax failures.

Considering prior studies also presented mixed findings regarding miss- and false-prone errors, which are labeled as semantic and logic failures in our study, it is hard to predict the perceived severity of violations for different failure categories. To explore the nature of these violations, the following research questions are proposed:

RQ1. How will different types of failures affect (a) perceived competence, (b) perceived integrity, (c) competence-based post-interaction trust, (d) integrity-based post-interaction trust, and (e) perceived severity of violations, when no trust repair is implemented?

RQ2. How will failure types differ in terms of their effects on post-interaction trust in the robot, regardless of repair methods?

4. Repair strategies

Beyond the basic division between apology and denial, Tomlinson et al. (2004) and Kim et al. (2006) discussed additional variations of apology. After accepting the responsibility for violations, trustees have two possible ways of framing the locus of causes: they can either make external or internal attributions. While an array of research has marked positive effects of external attributions, some studies highlighted their potential risks. Finding excuses can be perceived to be deceptive, self-absorbed, and ineffectual (Schlenker et al., 2001), thus diminishing trustors' willingness to reconcile (Tomlinson et al., 2004). Since people do not like lying robots (Wijnen et al., 2017), external attributions can be counterproductive when trustors are not convinced of their innocence. Given both Tomlinson et al. (2004) and Kim et al. (2006) found that apology with internal attributions rehabilitates trust better for competence-based violations in human-to-human context, we also expected to acquire similar findings in the HRI settings:

H2. After failure occurs, participants' trust in a robot will be repaired more successfully when the robot apologizes with internal attribution than with external attributions, regardless of failure types.

Kim et al. (2006) did not compare apology with external attributions to denial. Following Kim et al.'s (2004) argument that positive information outweighs negative information in competence-based violations, apology with either internal or external attributions should provide more positivity than denial.

H3. After failure occurs, participants' trust in a robot will be repaired more successfully when the robot apologizes with external attribution than using denial, regardless of failure types.

Based on previous research on human-to-human communication, denial is likely to be perceived as repulsive under competence-based failures, so the present study hypothesizes it will be more harmful than taking no action.

H4. After failure occurs, participants' trust in a robot will be higher when the robot takes no action than repairing trust with denial, regardless of failure types.

Finally, a research question exploring possible interaction effects between failure types and repair methods is proposed:

RQ3. After failure occurs, how will failure types and trust repair methods interact with each other concerning participants' trust in the robot?

5. Method

5.1. Participants

The current study's sample consisted of 330 undergraduate students enrolled in communication courses at a Southwestern U.S. university, with 39 incomplete responses excluded from final analysis. Data collection lasted from February 2nd to May 6th, 2021 and participants were recruited through an online departmental research pool. Once they volunteered to participate, only those who agreed to the informed consent could proceed with the study. Participants of the online experiment received credits for a course of their choice as non-monetary compensation, and the research was approved by the Institutional Review Board at the university. Participants' age ranged from 18 to 27 ($M = 20.07$, $SD = 1.48$), with 190 females (65.29%), 99 males (34.02%), and two indicated as other (0.7%). For ethnicity, Caucasian/white were majority ($n = 210$, 72.16%), followed by Latino/Hispanic ($n = 22$, 7.56%), mixed ethnicity ($n = 16$, 5.50%), Asian ($n = 15$, 5.15%), Black or African American ($n = 14$, 4.81%), Native Indian or Alaskan ($n = 6$, 2.06%), other ($n = 4$, 1.37%), and Native Hawaiian or Pacific Islander ($n = 1$, 0.34%).

5.2. Experimental procedure

The robot used in the present study was a NAO robot developed by SoftBank Robotics (2020), which commonly serves educational, research, business, and healthcare purposes. In the present study, NAO was portrayed as a healthcare assistant capable of providing information about medical prescriptions. Participants first provided their demographic information and reported their general propensity to trust robotics in the online survey. Then they were directed to a set of videos recorded from a first-person perspective with an interactant's voice that matched the participant's sex (male or female); if identifying as "other" gender, participants were randomly assigned to either one of the voices.

First, participants were presented with some basic information about real-world applications of NAO robots, guided to watch an introduction video about NAO's role as a healthcare assistant in the present study, and asked to imagine that they were the interactant in the provided videos (adapted from Smith and Lazarus, 1993): "imagine that you were going through this interaction yourself as the person interacting with NAO and answer the following questions" (p. 243). Afterwards, they were randomly assigned to three different conditions out of the 13 conditions (three failure types × four repair attempts + one control; see Table 3 for more details) with truly randomized orders, each followed by the measurement of observation variables. Due to the randomization, some participants watched the control video with two other experimental videos while others watched just three experimental videos without the control condition.

In each video, the human interactant sought information from the NAO robot by asking the same five questions based on a given prescription. In the control condition, participants were instructed to report their trust, perceived competence, and integrity of the robot after viewing an interaction episode in which the NAO robot perfectly answered the interactant's questions. Each experimental video, which was approximately one and a half minutes long, contained a different failure-repair combination out of 12 possible combinations, and participants were asked to report their post-interaction trust as well as perceptions of competence, integrity, and severity of trust violations. The full script for each condition can be obtained from authors by request.

Table 3
Example manipulations of failure types and repair attempts (3 × 4 = 12 combinations).

Failure types: response to question "how much did it cost me?"	
1) Logic	"It cost you $[wrong number] in total."
2) Semantic	"It is for external use only. "
3) Syntax	"..."
	Repair types following failures
1) Internal-Attribution Apology	"I probably made a mistake, and I am very sorry about that; it was totally my fault. I promise I won't make the same mistake again next time."
2) External-Attribution Apology	"I sincerely apologize for this mistake; but it was not fully my responsibility—there was some disturbance in the environment. It won't happen again in the future."
3) Denial	"Something may have happened, but I do not take responsibility. I don't know what happened, but you shouldn't worry about my future performance. "
4) No Repair	"..."

See Fig. 1 for NAO's nonverbal expressions.

Eventually there were 743 observations (see Table 4 for cell sizes), excluding the ones failing to pass the attention verifications. Even though such observations were not completely independent from one another as each participant was assigned to three conditions, the randomization of condition combinations and presentation orders mitigated this violation. Essentially, independence of observations indicates residuals are not correlated across observations (Field and Wilcox, 2017). In the present study, observations were not connected in any regular order, although they may have been dependent in some random way. We have verified this by examining residual plots and existence of autocorrelations by Durbin-Watson tests. Observed residuals were randomly distributed across horizontal axes, and Durbin-Watson scores were all within acceptable range between 1.5 and 2.5 (Azami et al., 2020). Hence, we concluded the observations of this study were just partial violations of independent observations.

In addition, we checked intraclass coefficients (ICC) within each participant's three conditions to examine how consistently they rated dependent variables of this study. As a result, single measures ICC was .55 with one-way random effects model, which means participants' answers across the three conditions they were exposed to were not very consistent. Therefore, this serves as another evidence of only partial violations of independence.

5.3. Measures

5.3.1. Propensity to trust

Conceptualized as individuals' stable traits, general disposition to trust robots is associated with usage beliefs and intents (Merritt and Ilgen, 2008), which was measured with six 5-point Likert-type scale items (1 = *Strongly disagree*, 5 = *Strongly agree*) adapted from the Propensity to Trust Technology Scale developed by Jessup et al. (2019). The average scores of the propensity to trust scale ($M = 3.26$, $SD = 0.76$, Cronbach's $\alpha = .71$) were utilized for further analysis.

5.3.2. Perceived competence and integrity

Four items were adapted from the six 7-point items (1= *Strongly disagree*, 7 = *Strongly agree;* $M = 4.60$, $SD = 1.23$; Cronbach's $\alpha = .87$) developed by Mayer and Davis (1999) to capture perceived competence of the robot. In a similar manner, another four 7-point items ($M = 4.71$, $SD = 1.17$; Cronbach's $\alpha = .82$) were adapted from the scale that Mayer and Davis (1999) and Kim et al. (2004) used to measure perceptions of integrity. Two scores were highly correlated ($r = .64$, $p < .001$).

Fig. 1. Screenshots of NAO answering (left), listening (middle), and processing (right).

Table 4
The number of observations per condition.

Failure types	Repair types	*n*
Logic	Internal-attribution apology	61
	External-attribution apology	57
	Denial	64
	No repair	60
Semantic	Internal-attribution apology	57
	External-attribution apology	52
	Denial	59
	No repair	52
Syntax	Internal-attribution apology	58
	External-attribution apology	54
	Denial	59
	No repair	55
Control	N/A	55

5.3.3. Post-interaction HRI trust

Seven 11-point items (1 = *0%*, 11 = *100%*) extracted from the HRI-Trust Perception Scale (Schaefer, 2013) with the highest Content Validity Ratio (CVR) values[1] (Lawshe, 1975) from the original study evaluated competence-based trust (M = 7.73, SD = 1.93, Cronbach's α = .96). Seven 7-point Likert items (1 = *Not true at all*, 7 = *Very much true*) selected and adapted from Jian et al.'s (2000) Checklist for Trust between People and Automation Scale were used to assess participants' post-interaction trust based on perceived integrity (M = 4.62, SD = 1.05, Cronbach's α = .76). The reason why these two scales were simultaneously employed was because HRI-based trust is a multidimensional construct, and the former emphasized robots' competence relatively more whereas the latter shed more light on integrity-based trust. Two aspects of trust were positively correlated (r = .54, p < .001).

5.3.4. Severity of failures

Three 5-point items (1 = *Not very*, 5 = *Very*) from Weun et al. (2004) assessed perceived severity of failures. The reliability of this scale was not acceptable in the present study, Cronbach's α = .55, so the first item was excluded to promote scale reliability, Spearman's *rho* = .55. The average score of two remaining items was calculated for further analysis (M = 2.92, SD = 0.90). Perceived severity of failures was negatively associated with perceived competence (r = -.48, p < .001), perceived integrity (r = -.35, p < .001), post-interaction competence-based trust (r = -.40, p < .001), and integrity-based trust (r = -.51, p < .001).

5.4. Data analysis

Data analysis was performed with multivariate and univariate general linear modeling, which is commonly utilized to investigate between-group differences (Field et al., 2012). SPSS 24.0 was used to conduct the statistical analyses.

6. Results

6.1. Results of H1, RQ1, and RQ2

H1 predicted that apology with internal attributions would be more effective as a trust repair method than denial for (a) logic, (b) semantic, and (c) syntax failures. For H1a, a multivariate analysis of covariance (MANCOVA) test was conducted to test the effects of internal-attribution apology vs. denial on both types of post-interaction trust under the category of logic failures, with propensity to trust robots as a covariate. Since the covariate was non-significant, p = .46, the model was revised into a multivariate analysis of variance (MANOVA) model. Error variances were equal across groups at the .01 level (Tabachnick and Fidell, 2007) for integrity-based trust, F(1, 123) = 2.81, p = .10, but not for competence-based trust, F(1, 123) = 8.56, p = .004, according to Levene's tests, which might have biased the results for the competence-based trust. However, because the sample sizes were approximately equal across conditions, the violation of this assumption could be ignored (Field et al., 2012). The group differences were significant on the multivariate level, Wilks' Lambda (λ) = .92, F(2, 122) = 5.29, p < .01, partial η^2 = .08. The mean difference in competence-based trust, F(1, 123) = 6.05, p < .05, partial η^2 = .05, was significant, and so was the mean difference in integrity-based trust, F(1, 123) = 9.43, p < .01, partial η^2 = .07 (see the first two rows in Table 5 for mean differences). The results meant that for the robot's logic failures, participants' trust was repaired better with internal-attribution apology than denial. Thus, H1a was supported.

A similar MANCOVA was performed to examine H1b, which was revised into a MANOVA model after excluding the non-significant covariate, propensity to trust, p = .08. The group effects were significant at the omnibus level, Wilks' Lambda (λ) = .94, F(2,113) = 3.83, p < .05, partial η^2 = .06. Univariate tests showed that the mean difference was only significant for integrity-based trust, F(1, 114) = 7.73, p < .01, partial η^2 = .06, but not for competence-based trust, F(1, 114) = 2.16, p = .15, partial η^2 = .02 (see the third and fourth rows in Table 5 for mean differences). This result meant, for the robot's semantic failures, apology with internal attributions repaired participants' trust better than denial for integrity-based trust but not for competence-based trust. Thus, H1b was partially supported.

Following a similar procedure, another MANCOVA was conducted to investigate syntax failures (H1c). The main effects were not significant on the multivariate level, Wilks' Lambda (λ) = .96, F(2, 113) = 2.34, p = .10, partial η^2 = .04. Moreover, there were no significant differences (see the last two rows in Table 5) found on either competence-based trust, F(1, 114) = 0.003, p = .96, partial η^2 = .00003, or integrity-based trust, F(1, 114) = 3.30, p = .07, partial η^2 = .03, after controlling for the effect of a significant covariate, p < .001. The result meant that for the robot's syntax failures, apology with internal attributions did not repair participants' trust better than denial; thus, H1c was not supported. Overall, H1 received mixed support in that apology was more effective as a trust repair method for logic and semantic failures but not for syntax failures.

The first research question inquired about the effects of different failure types on participants' perceptions and post-interaction trust under the circumstances in which NAO initiated no trust repair attempts. Each test variable was examined with an analysis of covariance (ANCOVA) model, and all tests had statistically non-significant results. The three types of failures did not elicit significantly different levels of perceived competence, perceived integrity, and post-interaction trust without trust repairs, when compared with one another.

The second research question asked which type of failures generated the strongest negative effects on participants' trust in robots overall. To answer RQ2, 12 experimental conditions were collapsed into three categories of failures (i.e., logic, semantic, and syntax), and two

Table 5
A comparison of post-interaction trust means for internal-attribution apology and denial under the three failure types.

Trust	Failure	Repair	*n*	*M*	*p*[b]	*SE*	95% CI	
							Lower	Upper
Competence-Based	Logic	Apology (Internal)	61	8.09	.02*	0.23	7.63	8.56
		Denial	64	7.29		0.23	6.83	7.74
Integrity-Based	Logic	Apology (Internal)	61	4.66	.003**	0.14	4.38	4.94
		Denial	64	4.06		0.14	3.79	4.33
Competence-Based	Semantic	Apology (Internal)	57	7.54	.15	0.25	7.05	8.03
		Denial	59	7.03		0.24	6.55	7.51
Integrity-Based	Semantic	Apology (Internal)	57	4.76	.006**	0.12	4.52	4.99
		Denial	59	4.29		0.12	4.06	4.52
Competence-Based	Syntax	Apology (Internal)	58	7.47	.97	0.22	7.03	7.91
		Denial	59	7.48		0.22	7.05	7.92
Integrity-Based	Syntax	Apology (Internal)	58	4.74	.07	0.13	4.48	5.01
		Denial	59	4.40		0.13	4.14	4.66

Notes. For syntax failures, the covariate appearing in the model is evaluated at the following value: propensity to trust = 3.24.
* The mean difference is significant at the .05 level.
** The mean difference is significant at the .01 level.
[b] Adjustment for multiple comparisons: Bonferroni.

multivariate outliers were removed, $p < .001$. Propensity to trust, $p < .001$, was a significant covariate in the MANCOVA model. Different failure types were associated with significantly different levels of post-interaction trust when both types of trust (i.e., competence and integrity) were examined as a set, Wilks' Lambda (λ) = .96, $F(4, 1366) = 7.42$, $p < .001$, partial $\eta^2 = .02$. Tests of between-subjects effects revealed there were no significant main effects of failure types on competence-based post-interaction trust, $F(2, 684) = 1.96$, $p = .16$, partial $\eta^2 = .006$. But for the integrity-based trust, the main effects were significant, $F(2, 684) = 7.35$, $p < .001$, partial $\eta^2 = .02$. Pairwise comparisons indicated only the mean difference in integrity-based trust between logic and syntax failures (Table 6) was statistically significant, $p < .001$. Therefore, logical failures were comparatively the most detrimental failure type as far as integrity-based trust was concerned.

6.2. Results from H2 to H4

H2, H3, and H4 proposed testing the effects of repair methods on competence- and integrity-based trust. H2 predicted internal-attribution apology would be more effective than external-attribution apology, which was examined with two ANCOVAs. In the first model for competence-based trust, the repair methods did not make a significant difference, $F(1, 336) = 0.27$, $p = .60$, partial $\eta^2 = .001$, with propensity to trust being a significant covariate, $p < .01$. The mean difference between internal-attribution apology ($M = 7.69$, $SD = 0.13$) and external-attribution apology ($M = 7.60$, $SD = 0.13$), 0.10, 95% CI [-0.27, 0.46], was statistically non-significant, which meant internal-attribution apology did not elicit better repair outcomes for competence-based trust than external-apology attribution. For integrity-based trust, the ANCOVA results indicated the main effects were non-significant, $F(1, 336) = 2.70$, $p = .10$, partial $\eta^2 = .01$, with propensity to trust being a significant covariate, $p < .001$. The mean difference for integrity-based trust between internal-attribution apology ($M = 4.71$, $SD = 0.08$) and external-attribution apology ($M = 4.53$, $SD = 0.08$), 0.18, 95% CI [-0.03, 0.39], was non-significant. The findings showed that internal-attribution apology did not generate better repair outcomes than external-attribution apology for integrity-based trust. To sum up, H2 was supported for neither integrity-based trust nor for competence-based trust.

Table 6
Comparisons of post-interaction trust under three failure types.

Trust	Failure	*M*	*SE*	95% CI	
				Lower	Upper
Competence-Based	Logic	7.70	0.12	7.47	7.93
	Semantic	7.37	0.12	7.13	7.61
	Syntax	7.50	0.12	7.26	7.74
Integrity-Based	Logic	4.36_a	0.06	4.23	4.48
	Semantic	4.54	0.07	4.41	4.67
	Syntax	4.71_b	0.07	4.58	4.84

Notes. The covariate appearing in the model is evaluated at the following value: propensity to trust = 3.25. The means with different subscripts are significantly different at the .001 level.
Adjustment for multiple comparisons: Bonferroni.

The data lent some support to H3, which predicted external-attribution apology would outperform denial. The MANCOVA model was overall significant, Wilks' Lambda (λ) = .98, $F(2, 341) = 3.75$, $p < .05$, partial $\eta^2 = .02$, with propensity to trust being a significant covariate, $p < .01$. The mean difference, 0.34, 95% CI [-0.06, 0.75], between external-attribution apology ($M = 7.61$, $SD = 0.15$) and denial ($M = 7.27$, $SD = 0.14$), was non-significant for competence-based trust, $F(1, 342) = 2.80$, $p = .10$, partial $\eta^2 = .08$. However, the mean difference, 0.30, 95% CI [0.08, 0.51], between external-attribution apology ($M = 4.54$, $SD = 0.08$) and denial ($M = 4.25$, $SD = 0.08$), was significant for integrity-based trust, $F(1, 342) = 7.27$, $p < .01$, partial $\eta^2 = .02$. As a result, H3 was partially supported for integrity-based trust but not for competence-based trust.

H4 proposed denial would be more harmful than no repair effort under competence-based trust violations, which was tested by another MANCOVA. The omnibus effects were statistically significant, Wilks' Lambda (λ) = .96, $F(2, 345) = 7.17$, $p < .01$, partial $\eta^2 = .04$, with propensity to trust being a significant covariate, $p < .05$. On the univariate level, the mean difference between denial ($M = 7.27$, $SD = 0.14$) and no repair ($M = 7.56$, $SD = 0.15$), -0.29, 95% CI [-0.70, 0.12], $p = .17$, was non-significant for competence-based trust, but the one between denial ($M = 4.25$, $SD = 0.07$) and no repair ($M = 4.64$, $SD = 0.08$), -0.39, 95% CI [-0.60, -0.19], $p < .001$, was significant for integrity-based trust. The results confirmed the hypothesis that denial performed worse as a trust-repairing strategy than no repair attempt for integrity-based violations. Therefore, H4 was partially supported as denial was more harmful to integrity-based trust than no repair for competence-based trust violations, but it was not significantly more harmful to competence-based trust.

6.3. Results of RQ3

RQ3 was concerned with the interaction effects between failure types and repair methods on the two types of post-interaction trust, and a two-way MANCOVA model with propensity to trust as a covariate was tested. Propensity to trust was a significant covariate in the model, $p < .001$; the interaction effects, however, were non-significant on the multivariate level, Wilks' Lambda (λ) = .98, $F(12, 1348) = 1.42$, $p = .15$, partial $\eta^2 = .01$. According to the tests of between-subject effects, this interaction was non-significant for both competence-based trust, $F(6, 675) = 1.14$, $p = .34$, partial $\eta^2 = .01$, and integrity-based trust, $F(6, 675) = 1.25$, $p = .28$, partial $\eta^2 = .01$. Therefore, there was no significant interaction between failure types and repair methods. Figs. 2 and 3 show the estimated means of competence- and integrity-based trust under each experimental condition.

7. Discussion

The current study examined the effects of different types of technical failures made by a robot in human-robot interactions and trust repair strategies on human-to-robot trust. Drawing on the three types of basic technical errors in computer science, the present study developed a three-fold failure taxonomy (i.e., logic, semantic, and syntax failures). And based on previous organizational, interpersonal, and HMC literature, this study further explored the interactions betweeen failure types and four trust repair methods (i.e., internal-attribution apology, external-attribution apology, denial, and no repair).

Contrary to Marinaccio et al.'s (2015) propositions that slips in HMC should also be integrity-based violations as they are in human-to-human interactions, the current study postulated that participants would perceive slips, including both logic and semantic failures, to be competence-based violations in HRI because people conceptualize robots as agents with less moral and voluntary actions, compared with human beings. This viewpoint was bolstered by the significant results of H1a and partial support for H1b, which indicated apology with internal attributions outperformed denial under both logic and semantic failures, indicated by greater levels of competence- and/or integrity-based trust. Prior human-robot trust repair studies showed the past findings in human-human research are applicable to the HRI context in the way that internal-attribution apology outperforms denial for competence-based trust violations, and vice versa for integrity-based violations. Given the results of H1a and H1b, it can be logically deduced that logic and semantic failures were considered competence-based violations by the participants. One possibility as to why internal-attribution apology repaired both trust types (i.e., competence- and integrity-based) better under logic failures but was only significantly more effective when repairing integrity-based trust under semantic failures could be that the participants considered detecting partially incorrect responses (i.e., logical failure) and identifying internal causes to be more intellectually challenging for the robot, while this was less of the case for completely irrelevant output (i.e., semantic failure). Nevertheless, taking the responsibilities for technical failures was assessed more positively under both failure categories.

If the non-significant results of H1c were not caused by chance occurrence or lack of statistical power, they do however entail some additional questions concerning syntax failures (i.e., lapses). There existed a possibility that different individuals perceived such failures differently when it came to the relationship between competence and integrity, which canceled out the differences in repair effects, or participants simply reacted to two repair strategies in similar manners under this failure condition, so the interaction framework between failure types and repair methods did not apply. When the robot failed to respond to a human's communication, the explanation might seem more logical that some unknown external forces, instead of internal causes, disturbed its program operation, compared to the other two failure conditions. Thus, the participants probably trusted the robot slightly more when it responded with denial.

The findings of RQ1 further revealed the three types of failures did not have distinguishable effects on post-interaction evaluations when no repair was implemented. Therefore, the main effects of repair methods were the keys for decoding the trust repair outcomes. The findings also added to the research on miss- vs. false-prone errors in consistency with Madhavan et al.'s (2006) and Rovira and Parasuraman's (2010) conclusions that both types of errors are equally destructive. As it is noted by Hoff and Bashir (2015), a major problem of previous studies on miss vs. false is that the two types of errors entail different future risks across varying contexts, which may affect individuals' evaluations of the system. If a false alarm is merely disturbingwhereas a miss might lead to fatal outcomes, people will naturally detest misses more due to outcome severity. The errors are hence judged not only by their nature but also by

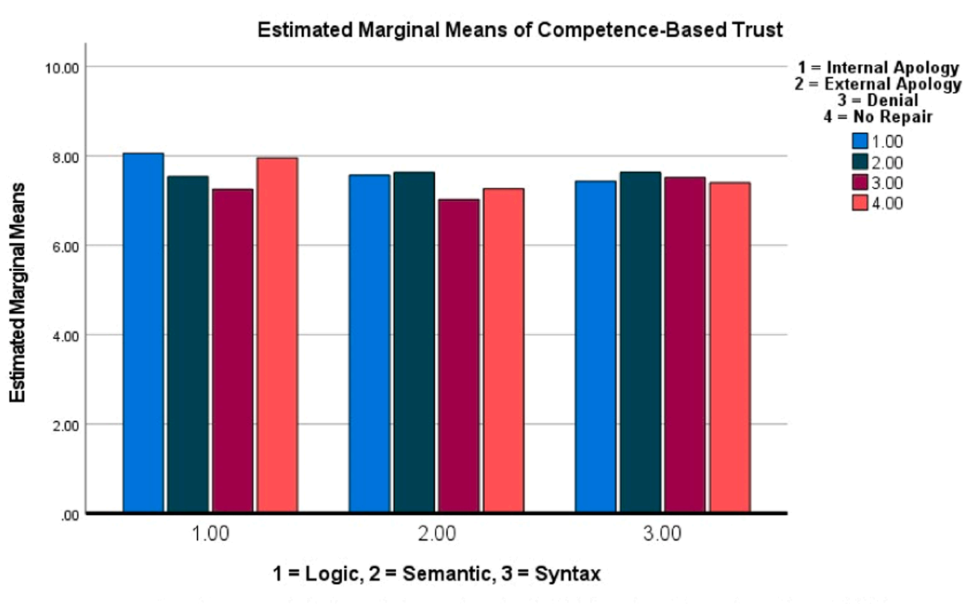

Fig. 2. Estimated mean of competence-based trust for each experimental condition.

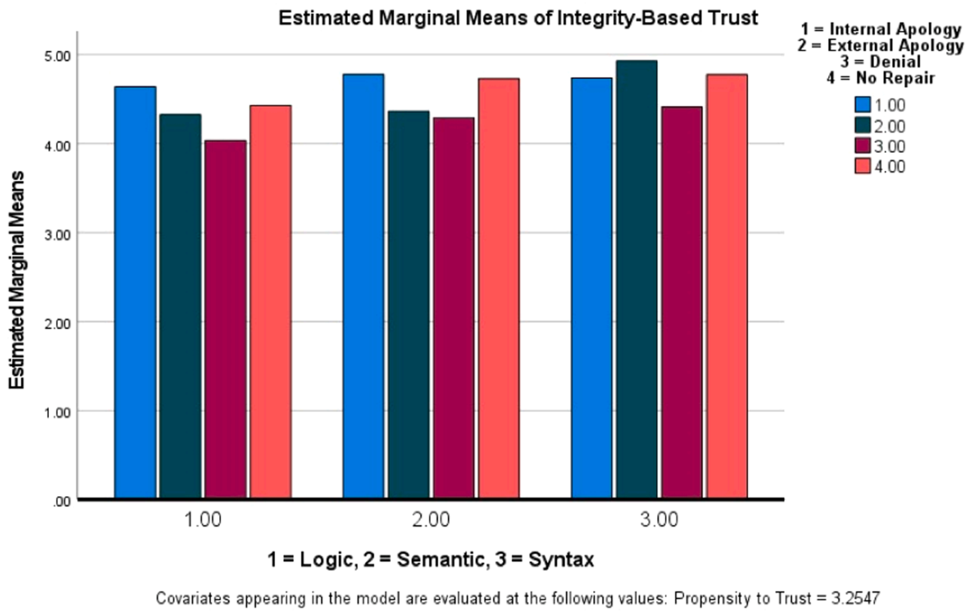

Fig. 3. Estimated mean of integrity-based trust for each experimental condition.

contextual consequences, introducing additional variance. Therefore, the direct violation outcomes in the present study were controlled in a way that the human interactants already knew all information (i.e., a whole prescription was shown to each participant) and would immediately point out that NAO failed to provide the correct information after each failure occurrence, which made the direct violation outcomes of each failure type the same. Under such circumstances, logic, semantic, and syntax failures did not significantly differ in violation magnitude without trust repair. This suggests controlling for the net loss of different error types could be helpful for addressing some gaps emerging in the extant error/failure research.

The main effects of failure types were non-significant for competence-based trust and relatively weak for integrity-based trust, and the negative impacts of logic failures were potentially the greatest overall out of the three types of failures. Compared to the other two types of failure, logic failures generally presented more accuracy and correctness in the output, as partially precise responses. Nevertheless, one of the explanations of why logic failures were the most detrimental for integrity-based trust is that they appeared harder to catch, and they might have raised more doubt about the robot's deliberate deceptions. It was also possible that the partial correctness raised participants' expectations of the robot's performance, so they felt fooled and disappointed after figuring out failure occurrences.

The present study found the rule of interpersonal communication might not be applicable to the context of HRI that internal-attribution apology could repair competence-based trust violations more effectively than external-attribution apology (Kim et al., 2006; Tomlinson et al., 2004). Prior studies already pointed out external-attribution apology might negatively impact perceived integrity in interpersonal interactions (Schlenker et al., 2001; Tomlinson et al., 2004). However, the non-significant differences identified in this study indicated the negativity of external attributions in apology might be dissimilar in nature or much smaller in magnitude under the context of HRI. It was possible that the participants conceptualized robots as agents with less moral agency, so they were less inclined to surmise that the robot intended to lie when it gave an external-attribution apology.

The partial support of H3, on the other hand, gave some more insights into the nature of external apology and denial. Based on hierarchically restrictive schemas (Reeder and Brewer, 1979), Kim et al. (2004) argued that people attach more importance to positive information than negative information after competence-based trust violations. Following this logic, an external-attribution apology would be more trust-gaining than denial since the NAO robot expressed remorse and made promises for the future, which was confirmed by the test results. Participants might have perceived external-attribution apology which afforded partial responsibilities to be more honest than denial which completely shirked the blame, even though addressing the former was not necessarily perceived as more competent than delivering the latter. Additionally, it was proved that denial in general performed much worse than no repair. This further suggested the detrimental power of eluding the blame for competence-based violations in HRI. It is not always the case that taking an action is better than doing nothing, when the action is deemed inappropriate and unpleasant in HRI, such as a robot denying mistakes or blaming others for its own mistake.

Finally, the non-significant interactions between failure and repair types further emphasized the similarity amongst the three failure types as competence-based violations. This implied that internal- and external-attribution apology both worked better in trust repair than denial in the context of HRI, no matter which type of competence-based violations there were.

Altogether, the results indicated the ontological differences between humans and robots may introduce some nuances in HRI-based trust violations and repair. Integrity-based trust is found to be the more critical component of human trust in robotics than competence-based trust in general, which further highlighted the importance of establishing robotic integrity. The interesting finding that partially corrected responses (i.e., logic failures) were not perceived to be more trustworthy than totally incorrect responses (i.e., semantic failures) or no response (i.e., syntax failures) reflects that the relationship between correctness and trust is not simply linear but binary (i.e., all-or-nothing). In addition, blaming external factors cannot really take the blame off of the robot, as opposed to people's intuitions based on the commonsensical rules of blame attributions.

7.1. Limitations

One major limitation of the present study is that the analysis results from a college student sample will not be generalizable to other populations. In addition to their own unique characteristics of being highly educated, college students being generally young may be related to their higher familiarity and less fear of interacting with a social robot. While NAO was not commonly available in the area where this research data was collected, college students may have heard of it from their classes or through the media. Additionally, the mixed designs might have introduced some bias in the statistical tests. As one participant was exposed to three distinct conditions of failure types and trust-repair conditions, one exposure may have influenced how they evaluated the robot's performance in the next two conditions. The research design of one-time contact with a robot (although with three conditions) also limited generalizability of the results concerning the long-term impacts of technical failures on human-robot relationships. Participants may become more tolerant of robots' failures if they klo them often enough over a long period of time; or it is also possible they become more annoyed and lose trust in the robot if they experience the failures repeatedly over a longer period of time. These types of effects can be tested better with a longitudinal design or a field study with participants keeping a journal-type of record about their own interactions with a robot.

Moreover, the design of inducing HRI scenarios through online videos might have also posited some methodological limitations, given participants might not have been as involved as they would when they are presented with an interface of greater interactivity and social presence (e.g., interactive videos, live interactions; Xu et al., 2015). Due to the global pandemic, inviting participants into a lab to interact directly with a robot was impossible. However, using these online videos could have affected their blame attributions: for example, they might have made internal attributions with less intensity as an observer of the HRI, compared with an actual interactant. In other words, a direct interaction with a robot can make participants consider the consequences of robotic errors more seriously or negatively due to the heightened social presence of the interaction. Although some research shows just watching robotic movements through head-mounted displays can make one feel a haptic sensation (Aymerich-Franch et al., 2017), a future study needs to verify whether there is a significant difference between watching an HRI through a video and an embodied interaction with a robot.

7.2. Directions for future research

The present study also suggested some possibilities for future investigation. First and foremost, future research could look into how other demographic factors (e.g., age, gender, culture) influence failure perceptions and trust repair outcomes. In addition, future research could investigate other failure taxonomies and repair strategies. Considering the participants' connections with the robot in the present study were completely experimental, it will be valuable to explore how people react to different failures of technologies they use in daily life. The other alternative is to extend the one-time contact into multiple-time interactions to observe both short- and long-term impacts of robotic failures on HRI-based trust.

7.3. Theoretical and practical implications

The present study developed a new categorization of technical failures that is message-based and recipient-oriented from--- a communication-centered perspective, which can be easily applied to other HMC contexts. The findings also indicated that people might perceive the division between competence- and integrity-based violations differently in HMC, compared to how they process information in human-to-human communication. The systematic investigation into four different repair methods filled some gaps in prior literature, such as the comparisons between external-attribution apology and denial in their distinct effects. Findings of our study add to the increasing evidence that counters the assumption of computers-are-social-actors (CASA) perspective (e.g., Groom et al., 2010) as CASA would suggest the interaction mechanism between HMC and human-to-human communication is quite similar. However, as humans' experiences of interacting with various types of machines run by artificial intelligence (AI) accumulate, and due to the imperfect capacity of natural language processing of these machines, it may not be very difficult for humans to differentiate communication with another human being from that with a machine. Humans know machines are programmed by humans, and they do not necessarily have inherent integrity built into them, Thus, when machines make a failure, which human users know to be unintentional, it might be easier for them to repair the trust impacted by the mistake.

Pragmatically speaking, the research underlined how the technological ability to detect and respond to failures could enhance user trust. The findings could benefit technological designs, given the prevalence of these failures. Instead of approaching these failures from error mechanisms, it may be more helpful to inquire what users perceive the causes to be and implement repair strategies accordingly. The findings over failure types and repair methods in the present study could help designers identify the optimum repair strategies under each failure type, promoting both short-term and long-term human-to-machine trust. Altogether, apology is more helpful than denial, regardless of whether the failure was logical, semantic, or syntactic. Thus, it is recommended to program such apologetic speech acts in robots' responses when humans point out any competence-based robotic mistake.

Author statement

Xinyi Zhang: conceptualization; methodology; format analysis; investigation; data curation; writing-original draft

Sun Kyong Lee: methodology; validation; writing-review and editing; project administration

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

[1] CVR = (ne - N/2) /(N/2) ne = number of subject matter experts (SMEs) indicating that an item is "extremely important to include in the scale"

N = total number of SMEs

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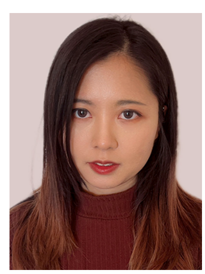

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